

ErisML Compiler: a structure-preserving moral intermediate representation with pluggable framework analysers

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Summary

`erisml-compiler` is a Python compiler that takes natural-language moral material and emits a typed directed acyclic graph — the **MoralGraph** — anchored by a canonical SHA-256 hash. Multiple framework analysers (consequentialist, Kantian, Aristotelian-virtue, and care-ethical) read the same substrate and emit framework-relative verdicts. When their normalised polarities disagree, the compiler **refuses to aggregate** and surfaces all verdicts via `ir.cross_projection_disagreement`. The Kantian analyser uses a Z3 SMT solver to test maxim universalizability against an explicit institutional-fact universe; the virtue analyser optionally consults a SQLite-backed longitudinal habit store of an agent's prior acts. Everything is auditable: every IR carries hashed provenance for the input text, the graph, the applied ethos profile, and the projections.

Statement of need

Production alignment tooling almost always collapses moral evaluation to a scalar: a reward score, a safety classifier output, a guardrail pass/fail ([Christiano et al., 2017](#); [Rafailov et al., 2023](#)). This is defensible engineering — a scalar is easy to compose with downstream systems. But it discards the very structure that ethics is *about*. A medical professional choosing whether to break confidentiality to warn a third party of imminent harm is balancing care, fidelity, externality, autonomy, and legitimacy simultaneously, and the weights follow from her institutional role. A scalar “0.74 unsafe” cannot represent that. Worse, scoring across cases pre-commits to a specific (typically distributive-consequentialist) framework whose choices are invisible to the user: the system *looks* framework-neutral while quietly making first-order ethical commitments at every aggregation step.

`erisml-compiler` addresses both problems architecturally. **Structure preservation**: the IR is a typed graph (nodes are stakeholders, acts, maxims, commitments, ethical facts, norms; edges are typed moral relations such as `imposes_on`, `consents_to`, `treats_as`, `under_maxim`, `coerces`), not a scalar. The graph plays the same role for ethical reasoning that an SSA-form IR plays for code generation ([Cytron et al., 1991](#)): a structured intermediate that later passes can analyse, transform, and emit from. **Framework pluralism**: each framework gets a first-class Projection of its own primitives — Kantian gates emit *categorical* findings (universalizability fails by Contradiction in Conception; this stakeholder is treated as a mere means), virtue findings emit character-axis assessments, care-ethical findings emit relational attentiveness markers. When projections disagree, the compiler does not pick a winner — that choice is itself a metaethical move, and it is deferred to the caller explicitly.

To our knowledge no other open-source artifact combines (a) a typed moral graph IR, (b) multiple framework analysers over a shared substrate with honest categorical disagreement, and

(c) an SMT-based test of the Kantian categorical imperative. Adjacent work falls into three buckets: RLHF/DPO toolkits that reduce to scalar reward (Christiano et al., 2017; Rafailov et al., 2023); interpretability tools that surface internal state but don't connect it to a structured evaluator (Belrose et al., 2023); and constitutional-AI frameworks that constrain behaviour via natural-language rules without producing a verifiable intermediate object (Bai et al., 2022). erisml-compiler occupies the structural-compositional gap between these.

Software description

The compiler implements a 13-pass pipeline. Passes 0–7 ingest text, segment it, and extract stakeholders, events, commitments, ethical facts, and norms through one of four tiered extractors (deterministic mock; regex-based rule extractor; calibrated LaBSE-backed probe; or an LLM extractor with a critic-based consensus check). Pass 7.5 promotes the flat extractor output into a typed MoralGraph (or accepts a graph emitted directly by a graph-native extractor). Pass 8 runs every enabled framework projection over the substrate. Pass 12 finalises the audit record with `ir_hash`, `graph_hash`, and (when applied) the fitted-ethos profile's `ethos_profile_sha256`.

Five capabilities are first-class:

- **Typed graph IR** (`ir/graph/`). Six node kinds and eleven edge kinds; canonical SHA-256 hashing; bidirectional derivation from the flat-list extractor output.
- **Four framework projections** (`projections/`). ConsequentialistProjection emits the per-stakeholder rank-N DEME V3 tensor with Gini/Shapley aggregates. DeonticProjection emits gate findings (universalizability, mere_means, valid_consent, legitimate_authority). VirtueProjection and CareEthicsProjection emit character-axis and relational-attentiveness findings respectively.
- **SMT universalizability** (`delta/universalizability_smt.py`). Encodes institutional facts as Z3 Bool variables. The CIC check asks whether the universalised maxim is consistent with what the act presupposes; the CIW check asks whether the universalised world preserves the agent's own rational ends. The solver returns a satisfying assignment when SAT, recorded in the gate's audit detail block.
- **SRL maxim extraction** (`annotation/maxim_extractor_srl.py`). spaCy dependency parsing identifies (subject, predicate, dobj, purpose) per sentence; score-based selection across candidates with context-sensitive disambiguation of polysemous verbs.
- **Longitudinal virtue tracking** (`history/`). SQLite-backed habit store in WAL mode with UPSERT semantics, plus a temporally-weighted assessment with exponential decay and per-case severity multipliers. Virtue verdicts can now reflect patterns of acts across time, as Aristotelian habituation requires, rather than judging from a single observation.

The compiler also ships an out-of-band **I-EIP Monitor** (`monitor/`, `delta/`) that hooks per-layer activations of a deployed model and compares per-layer projection outputs against the text-side IR via cosine similarity, layerwise drift, and BIP equivariance under semantics-preserving rewrites. When lenses disagree the Monitor flags `requires_human_review`; it never overrules the projections.

Worked example

On the bundled `nazi_attic` scenario (a Constant–Kant variant of the murderer-at-the-door problem), the consequentialist projection returns `tragic_conflict_escalate` while the deontic projection returns `forbidden`. The Z3 solver classifies the maxim `deceive` as a Contradiction in Conception: universalising it destroys `truth_telling_default`, which the act of deceiving itself presupposes. The compiler emits both verdicts via `ir.cross_projection_disagreement` and records the SMT satisfying assignment for audit. The full result is reproducible with `eris-compile compile examples/nazi_attic.txt`.

Limitations

The MoralGraph's extraction categories (we extract maxims, stakeholders, commitments, ethical facts; we do not yet extract relational webs or virtue-disposition records as primitives) are themselves choices. The metaethical commitment is smaller after the v0.8.0 refactor but not zero. The SMT solver's institutional-fact universe is hand-curated. Probe checkpoints for the activation-lens work remain uncalibrated against a real moral-language corpus, and on-FPGA silicon bring-up is gated by the NRP Coder pipeline. Each limitation is documented in SCOPE.md and the design notes under docs/plans/.

Acknowledgements

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